ECE 310 Fall 2026
Lecture 11: BIBO stability and causality

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Learning Objectives

- Explain the conditions for an LTI system to be BIBO stable using: the definition of BIBO stability, impulse response, or z-transform.
- Define what makes a causal and non-causal LTI systems in the z-domain.
- Identify bounded inputs that cause unstable systems to yield unbounded outputs.



BIBO Stability

Thus far, we have given two conditions for BIBO stability of a discrete-time system:

1. Following the definition of BIBO stability: A discrete-time system T is *bounded-input bounded-output stable* (BIBO stable) if and only if for any $x[n]$ such that $|x[n]| < \beta$ for all n, $|T(x[n])| < \alpha$ for all n where $0 \leq \alpha < \infty,\ 0 \leq \beta < \infty$.

2. For an LTI system with impulse response $h[n]$: An LTI system is BIBO stable if and only if

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty.$$



BIBO Stability in the z-domain

We can also demonstrate another necessary and sufficient condition for BIBO stability in the z-domain. Let $H(z)$ be the transfer function of an LTI system with an ROC that contains the unit circle, i.e. $|z| = 1$. This implies:

$$\begin{aligned}
\infty &> \left(\sum_{n=-\infty}^{\infty} |h[n]z^{-n}| \right) \Big|_{z=e^{j\theta}} \\
&= \sum_{n=-\infty}^{\infty} |h[n]e^{-j\theta n}| \\
&= \sum_{n=-\infty}^{\infty} \left| \sqrt{(h[n]e^{-j\theta n})(h^*[n]e^{j\theta n})} \right| \\
&= \sum_{n=-\infty}^{\infty} |h[n]|.
\end{aligned}$$



Relating BIBO Stability, Causality, and z-domain

Recall that infinite-length signals can fall into one of three categories:

1. Right-sided: $x[n] = 0$, $n < n_0$, $n_0 \in \mathbb{Z}$
   - The ROC of right-sided signals is $|z| > a$, $a > 0$.
2. Left-sided: $x[n] = 0$, $n > n_0$, $n_0 \in \mathbb{Z}$
   - The ROC of left-sided signals is $|z| < b$, $b > 0$.
3. Two-sided: $x[n]$ is nonzero for $-\infty < n < \infty$.
   - The ROC of two-sided signals is $a < |z| < b$, $0 < a < b$.



Relating BIBO Stability, Causality, and z-domain: Right-sided Signals

- Right-sided: $x[n] = 0$, $n < n_0$, $n_0 \in \mathbb{Z}$
  - The ROC of right-sided signals is $|z| > a$, $a > 0$.



Relating BIBO Stability, Causality, and z-domain: Left-sided Signals

- Left-sided: $x[n] = 0$, $n > n_0$, $n_0 \in \mathbb{Z}$
  - The ROC of left-sided signals is $|z| < b$, $b > 0$.

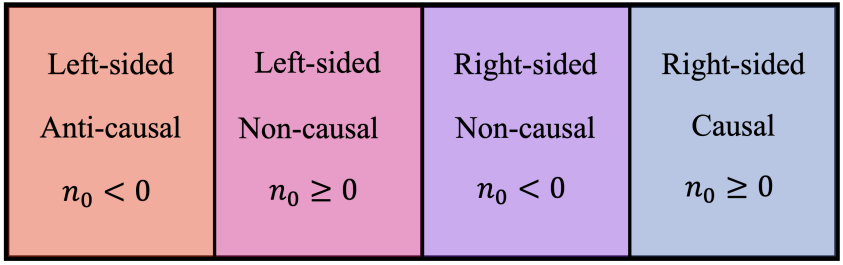


Relating BIBO Stability, Causality, and z-domain: Two-sided Signals

- Two-sided: $x[n]$ is nonzero for $-\infty < n < \infty$.
  - The ROC of two-sided signals is $a < |z| < b$, $0 < a < b$.



Relating BIBO Stability, Causality, and z-domain: Summary



ROC shape of $H(z)$	Causality Type	Right-sided	Left-sided	Condition for stability
$\|z\| > \|p_{\max}\|$	Causal	✓	✗	$\|p_{\max}\| < 1$
$\|z\| < \|p_{\min}\|$	Anti-causal	✗	✓	$\|p_{\min}\| > 1$
$\|p_{\max}\| < \|z\| < \infty$	Non-causal	✓	✗	$\|p_{\max}\| < 1$
$0 < \|z\| < \|p_{\min}\|$	Non-causal	✗	✓	$\|p_{\min}\| > 1$
$a < \|z\| < b$	Non-causal	✓	✓	$\|p_{\max,h_r}\| = a < 1$ and $\|p_{\min,h_l}\| = b > 1$

Example: BIBO Stability of IIR Systems

For each LTI system, determine the corresponding $h[n]$ for which the system is BIBO stable. Also state if this stable system is causal, non-causal, or anti-causal.

$$H(z) = \frac{1}{1 - \frac{1}{2}z^{-1}} + \frac{1}{1 + \frac{4}{3}z^{-1}}$$

$$H(z) = \frac{1}{1 - \frac{1}{5}z^{-1}} + \frac{1}{1 - \frac{4}{5}z^{-1}}$$



Example: FIR System Stability

Can an LTI FIR system be unstable? Justify your reasoning in the time-domain and z-domain.



Unstable Systems and Unstable Inputs

When a system is unstable, it is important to identify what (bounded) input signals may lead to an unbounded output. Consider, for example, an unstable LTI system with impulse response $h[n] = 3^n u[n]$. What bounded inputs yield unbounded outputs for this system?



Unstable Systems and Unstable Inputs

Consider, for example, an unstable LTI system with impulse response $h[n] = 3^n u[n]$. What bounded inputs also yield bounded outputs for this system?



Unstable Systems and Unstable Inputs

Suppose instead we have an unstable LTI system with the following impulse response.

$$h[n] = (-1)^n u[n]$$



Marginal Stability

Marginal Stability

An LTI system with impulse response $h[n]$ and transfer function $H(z)$ is *marginally stable* if its ROC is open on the unit circle, i.e. $|z| < 1$ or $|z| > 1$.

What bounded inputs produce unbounded outputs for marginally stable systems?



Pole Matching: Time-Domain Interpretation

A causal marginally stable system can be described by an impulse response $h[n] = a^n u[n]$, for $|a| = 1, a \in \mathbb{C}$. Let $x[n] = b^n u[n]$. When is $y[n] = x[n] * h[n]$ an unbounded output?

Three cases to consider:

(a) $|b| > 1$

(b) $|b| < 1$

(c) $|b| = 1$



Pole Matching: Time-Domain Interpretation

A causal marginally stable system can be described by an impulse response $h[n] = a^n u[n]$, for $|a| = 1, a \in \mathbb{C}$. Let $x[n] = b^n u[n]$. When is $y[n] = x[n] * h[n]$ an unbounded output?

(c) If $|a| = 1$ and $|b| = 1$, then let $a = e^{j\theta}$ and $b = e^{j\phi}$:



Example: Pole Matching

Consider a causal LTI system with the following transfer function:

$$H(z) = \frac{1}{(1 - jz^{-1})(1 + z^{-1})(1 - \frac{2}{3}z^{-1})}.$$

Which of the following input signals will lead to an unbounded output?

1. $(-1)^n u[n]$
2. $u[n]$
3. $e^{j\frac{\pi}{2}n} u[n]$
4. $e^{-j\frac{\pi}{2}n} u[n]$
5. $\left(\frac{2}{3}\right)^n u[n]$
6. $\sin\left(-\frac{\pi}{2}n\right) u[n]$
7. $\cos(\frac{\pi}{2}n) u[n]$
8. $\sin\left(\frac{2\pi}{3}n\right) u[n]$